

ControlFace: Harnessing Facial Parametric Control for Face Rigging

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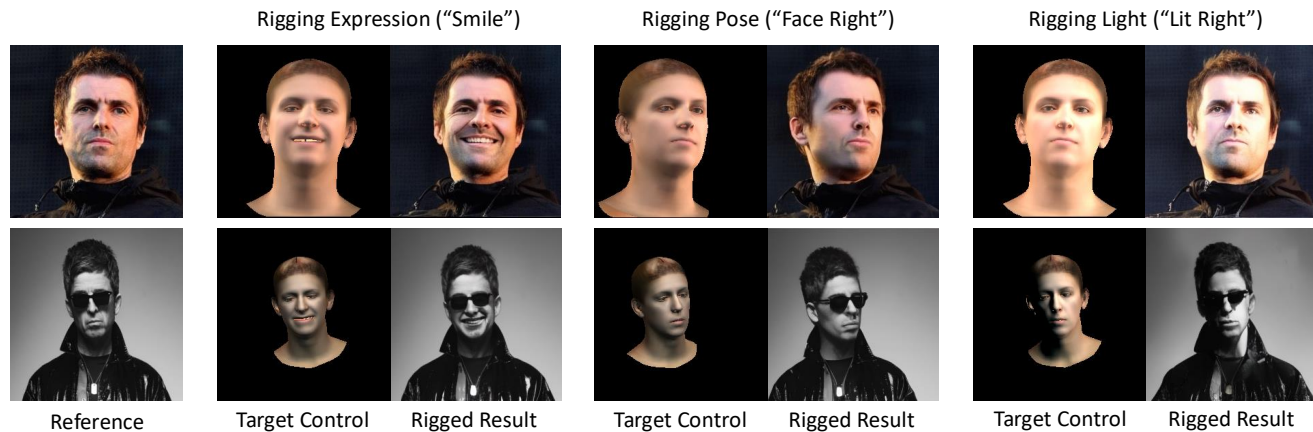


Figure 1. Our **ControlFace** can edit the input face image using explicit facial parametric controls, generating realistic images without compromising the identity and other semantic details such as hairstyle.

Abstract

Manipulation of facial images to meet specific controls such as pose, expression, and lighting, also known as face rigging, is a complex task in computer vision. Existing methods are limited by their reliance on image datasets, which necessitates individual-specific fine-tuning and limits their ability to retain fine-grained identity and semantic details, reducing practical usability. To overcome these limitations, we introduce ControlFace, a novel face rigging method conditioned on 3DMM renderings that enables flexible, high-fidelity control. We employ a dual-branch U-Nets: one, referred to as FaceNet, captures identity and fine details, while the other focuses on generation. To enhance control precision, the control mixer module encodes the correlated features between the target-aligned control and reference-aligned control, and a novel guidance method, reference control guidance, steers the generation process for better control adherence. By training on a facial video dataset, we fully utilize FaceNet’s rich representations while ensuring control adherence. Extensive experiments demonstrate ControlFace’s superior performance in identity preservation and control precision, highlighting its practicality. Please see the project website: <https://cvlab-kaist.github.io/ControlFace/>.

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1. Introduction

Manipulating input facial images has long been a fundamental task in computer vision and graphics [2, 26, 43]. Given a face image, the goal of face rigging is to generate an image that corresponds to user controls—lighting or facial pose—without compromising the identity of the face. This task is often complicated by the need to preserve fine details like hairstyle, making it a challenging problem.

Recent advancements in generative models [8, 26, 33, 34, 41] have significantly impacted the development of face rigging techniques. Diffusion models, in particular, have demonstrated remarkable capabilities in producing high-quality, realistic facial images through their iterative refinement processes [17, 41]. These models have shown promise in generating realistic facial images. Leveraging 3D face reconstruction methods [4, 9, 54], enabling the estimation of 3D Morphable Model (3DMM) parameters—such as those of FLAME [24]—directly from an image, many works [7, 10, 19, 25, 28, 42] have utilized 3DMM parameters or renderings as target controls for face rigging. However, existing methods still face several limitations, including difficulties in handling real images [10, 42], the requirement for fine-tuning on images of the same identity [7, 19], and challenges in maintaining fine-grained details from reference images [25, 28]. Addressing these issues is crucial

ControlFace:利用面部参数控制实现面部绑定

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图1. 我们的ControlFace能够通过显式面部参数控制对输入人脸图像进行编辑，生成逼真图像，同时不损害身份特征及其他语义细节（如发型）。

摘要

在计算机视觉领域，对人脸图像进行姿态、表情和光照等特定参数的操控（即面部绑定技术）是一项复杂任务。现有方法受限于对图像数据集的依赖，需要针对个体进行精细调整，导致难以保留细粒度的身份特征和语义细节，实用性大打折扣。为突破这些局限，我们提出基于3DMM渲染的新型面部绑定方法ControlFace，实现灵活且高保真度的控制。该方法采用双分支UNet架构：其中分支FaceNet负责捕捉身份特征和细节信息，另一分支则专注于图像生成。为提升控制精度，控制混合模块通过编码目标对齐控制与参考对齐控制之间的关联特征，并结合创新的参考控制引导方法，优化生成过程以增强控制一致性。通过在面部视频数据集上训练，我们充分挖掘了FaceNet的丰富表征能力，同时确保了控制效果的稳定性。大量实验表明ControlFace在身份保持和控制精度方面具有卓越性能，凸显其实用价值。请访问项目官网：<https://cvlab.kaist.github.io/ControlFace/>。

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1. 介绍

对输入的面部图像进行处理一直是计算机视觉与图形学领域的基础任务[2, 26, 43]。给定一张面部图像，面部绑定的目标是在不破坏面部特征的前提下，生成与用户控制（如光照或面部姿态）相对应的图像。由于需要保留发型等细微特征，这项任务往往变得复杂，因此成为一个具有挑战性的难题。

生成模型的最新进展[8, 26, 33, 34, 41]显著影响了面部绑定技术的发展。特别是扩散模型，通过其迭代优化过程[17, 41]展现出生成高质量、逼真面部图像的卓越能力。这些模型在生成逼真面部图像方面展现出巨大潜力。借助三维面部重建方法[4, 9, 54]，可直接从图像中估算三维可变形模型（3DMM）参数——例如火焰参数[24]——许多研究[7, 10, 19, 25, 28, 42]已将3DMM参数或渲染结果作为面部绑定的目标控制。然而，现有方法仍面临若干局限性，包括处理真实图像的困难[10, 42]、需要对同一身份图像进行微调的要求[7, 19]，以及从参考图像中保持精细细节的挑战[25, 28]。解决这些问题至关重要。

to enhance the practicality and robustness of face rigging approaches in real-world applications.

Notably, conventional diffusion-based face rigging methods [7, 19, 25, 28] rely on single-image-per-individual datasets such as FFHQ [20]. Consequently, these methods are inevitably trained in a reconstruction manner, where only the reference image and its aligned control are available. Specifically, the reference image is embedded through an encoder and the model learns to reconstruct the input reference image with the embedded feature and the aligned control. However, if the feature from the reference image conveys too much information, reconstruction task becomes trivial, which often leads the model to ignore the control signal and simply replicate the reference image making rigging not possible as shown in Fig. 2. Hence, to make the reconstruction non-trivial, they often encode the reference image into a single vector using a network such as pretrained face recognition model [5]. This compression into a single vector, however, may discard essential facial details failing to capture both identity and details simultaneously. This leads to additional individual specific fine-tuning [7, 19]. However, fine-tuning requires images with the same identity as the reference image which may be cumbersome to obtain.

In this paper, we introduce **ControlFace**, a novel face rigging method conditioned on 3DMM renderings, which effectively integrates both target control and reference control extracted from the reference image. Unlike previous methods [7, 10, 25], ControlFace enables flexible edits to lighting, shape, expression, and pose without requiring fine-tuning, while preserving both identity and semantic details of the input reference image. To achieve this, we use dual-branch U-Nets initialized from a large pretrained model: one, termed *FaceNet*, which encodes the reference image to capture its identity and semantic details, and the second *denoising U-Net* that handles the generation. The rich representation encoded by the FaceNet is concatenated with the intermediate feature of the denoising U-Net and integrated through self-attention layer. For better control adherence we introduce the control mixer module (CMM), which embeds the correlated features between the target and reference controls. Additionally, we introduce a reference control guidance (RCG) mechanism that uses the reference control as a null condition label to improve grounding, inspired by Classifier-Free Guidance (CFG) [16]. ControlFace is trained on a facial video dataset [52], which provides paired reference images and target images aligned with the desired control. This training approach allows us to avoid reconstruction-based training and hence fully leverage FaceNet’s rich generative representation without ignoring the target control.

We conduct extensive experiments and ablation studies to rigorously evaluate the effectiveness of our proposed method. We demonstrate ControlFace’s superior perfor-



Figure 2. **Limitations of reconstruction-based training.** We compare the results of our model trained on an image dataset [20] in a reconstruction setup and on a video dataset [52] with paired samples created by randomly selecting two frames from each video. The results by reconstruction-based training often ignores the target control at inference.

mance in identity preservation, semantic consistency, and control precision compared to existing methods, by evaluating on multiple unseen faces [20]. We also show that ControlFace outperforms fine-tuning-based methods [7] without the need for the cumbersome additional training. Ablation studies further highlight the distinct contributions of all the components in our model. These results validate our approach and illustrate its potential for real-world applications in face rigging.

2. Related Work

Generative Face Editing. Since the introduction of GANs [11], face generation has received significant attention. Early GAN-based methods [13, 39, 44] leveraged the disentangled latent space of StyleGAN [20] to manipulate facial attributes by finding specific directions in latent space. Other approaches [6, 10, 26, 38, 42] incorporated 3DMMs [2, 3] to conditionally control pose and expression through 3D structural guidance, improving realism and consistency.

Recently, diffusion models have emerged as a powerful alternative for generative face editing due to their stable training and high-quality output. For identity-preserving generation and manipulation, recent diffusion-based works either incorporate facial features directly [12, 25, 45, 46, 48], train networks to encode images [7, 19, 30, 49], or leverage pretrained diffusion U-Nets for rich representations [18, 47, 53].

Controllable Generation for Diffusion Models. Methods for conditioning diffusion models with spatial controls initially relied on concatenation [34–36]. The advent of large-scale pretrained text-to-image models [8, 34] has leveraged sophisticated text embeddings [31, 32] processed through advanced attention mechanisms, driving research into finer control over the generation process [23, 27, 29,

以提升面部绑定方法在实际应用中的实用性和稳健性。

值得注意的是，传统的基于扩散的面部绑定方法[7, 19, 25, 28]依赖于单个体单图像数据集，例如FFHQ[20]。因此，这些方法不可避免地以重建方式训练，仅能获得参考图像及其对齐的控制信号。具体而言，参考图像通过编码器嵌入，模型学习利用嵌入特征和对齐控制信号重建输入参考图像。然而，若参考图像的特征传递过多信息，重建任务将变得简单，这常导致模型忽略控制信号，直接复制参考图像，如图2所示，从而无法实现绑定。因此，为使重建任务复杂化，通常会使用预训练人脸识别模型[5]等网络将参考图像编码为单一向量。但这种向量压缩可能丢失关键面部细节，无法同时捕捉身份特征与细节信息。这导致需要额外的个体特异性微调[7, 19]。然而，微调需要与参考图像具有相同身份的图像，这可能难以获取。

在本文中，我们介绍了**ControlFace**，一种基于3DMM渲染的新型面部绑定方法，该方法有效整合了从参考图像中提取的目标控制和参考控制。与先前的方法[7、10、25]不同，ControlFace能够在无需微调的情况下灵活编辑光照、形状、表情和姿态，同时保留输入参考图像的身份和语义细节。为实现这一点，我们使用从大型预训练模型初始化的双分支U-Net：一个称为FaceNet，用于编码参考图像以捕捉其身份和语义细节；另一个是处理生成的去噪U-Net。FaceNet编码的丰富表征与去噪U-Net的中间特征相拼接，并通过自注意力层进行整合。为增强控制一致性，我们引入了控制混合模块（CMM），该模块嵌入了目标控制与参考控制之间的相关特征。此外，我们引入了一种参考控制引导（RCG）机制，该机制将参考控制作为无效条件标签来增强定位精度，其灵感源自无分类器引导（CFG）[16]。ControlFace模型基于面部视频数据集[52]进行训练，该数据集包含与目标控制对齐的配对参考图像和目标图像。这种训练方法使我们能够避免基于重建的训练，从而充分利用FaceNet丰富的生成表征，同时不忽略目标控制。

我们通过大量实验和消融研究，严格评估了所提出方法的有效性。实验结果表明，ControlFace在...



参考对照 结果（Rec.）结果（Ours）

图2. **基于重建的训练的局限性。** 我们将基于重建设置的图像数据集[20]训练的模型结果与基于视频数据集[52]训练的模型结果进行比较，后者通过从每个视频中随机选取两帧创建配对样本。基于重建的训练结果在推理时经常忽略目标控制。

通过在多个未见过的脸数据集上进行评估[20]，我们发现ControlFace在身份保持、语义一致性和控制精度方面均优于现有方法。研究还表明，ControlFace无需进行繁琐的额外训练，即可超越基于微调的方法[7]。消融实验进一步凸显了模型中各组件的独特贡献。这些结果不仅验证了我们的方法，更展现了其在面部绑定实际应用中的潜力。

2. 相关工作

生成式人脸编辑。 自生成对抗网络（GAN）问世以来[11]，人脸生成技术便备受关注。早期基于GAN的方法[13、39、44]利用StyleGAN[20]的解耦潜在空间，通过在潜在空间中寻找特定方向来操控面部特征。其他方法[6、10、26、38、42]则采用三维蒙特卡洛方法[2、3]，通过三维结构引导条件控制姿态与表情，从而提升真实感与一致性。

最近，扩散模型因其稳定的训练过程和高质量的输出，已成为生成式人脸编辑的强大替代方案。在身份保持生成与操作方面，近期基于扩散的研究要么直接整合面部特征[12, 25, 45, 46, 48]，要么训练网络对图像进行编码[7, 19, 30, 49]，要么利用预训练的扩散U-Nets实现丰富表征[18, 47, 53]。

扩散模型的可控生成。 初始阶段，通过空间控制条件化扩散模型的方法依赖于拼接[34–36]。大规模预训练文本到图像模型的出现[8, 34]利用了通过高级注意力机制处理的复杂文本嵌入[31, 32]，推动了对生成过程进行更精细控制的研究[23, 27, 29]，

50]. Recently, ControlNet [50] has enabled precise control over attributes like depth, edges, and poses by integrating timestep-specific control features into the diffusion U-Net. In face editing, approaches [7, 12, 19, 25] have adopted similar architectures with 3DMMs to allow explicit facial control. However, these methods often struggle to encode fine details from the reference image. To overcome this limitation, we employ dual-branch structure: one for encoding the reference facial image and the other for generating manipulated image.

3. Preliminaries

3D Morphable Face Models (3DMMs). 3DMMs [2, 3, 24] are facial parametric models designed to accurately represent facial pose, shape, and expression using a compact latent. FLAME [24] which is one of the widely used 3DMM, takes shape, pose, and expression as inputs and outputs a face mesh. In this work, we leverage DECA [9] model which is capable of predicting the FLAME [24] parameters together with the orthographic camera parameters, albedo, and lighting in spherical harmonics from a facial image. With these parameters we can apply Lambertian reflectance to acquire surface normals, albedo, and Lambertian rendering. These renderings offer pixel-aligned control, providing both geometric and texture information that suitable for face rigging. Hence, we leverage these renderings as our control.

Latent Diffusion Models (LDM). LDM [34] is a generative model that progressively learns to denoise random Gaussian noise in the latent space of VAE [21]. During training, an input image X is encoded into a latent representation z via a VAE encoder. This latent z is then perturbed with Gaussian noise ϵ through a predefined diffusion forward process at a randomly selected timestep t , producing the noisy latent z_t . The model’s core objective is to predict ϵ , effectively learning to reverse the noise. The training objective can be written as:

$$\mathcal{L}_{LDM} = \mathbb{E}_{z,t,\epsilon \sim \mathcal{N}(0,1)} [\|\epsilon_\theta(z_t; t) - \epsilon\|_2^2]. \quad (1)$$

Classifier-Free Guidance (CFG). To enhance conditional generation of diffusion model during inference, CFG [16] reformulates the predicted noise steering the generation process away from the unconditional distribution and towards the conditional distribution. This allows us to sample from a sharper conditional distribution, enhancing the condition fidelity. The modified prediction can be formulated as follows:

$$\hat{\epsilon}_\theta(\cdot, C) = \epsilon_\theta(\cdot, \emptyset) + w(\epsilon_\theta(\cdot, C) - \epsilon_\theta(\cdot, \emptyset)), \quad (2)$$

where $\emptyset$ denotes an unconditional label such as empty text, C denotes the conditional input and w is a guidance scale. Hence, $\epsilon_\theta(\cdot, \emptyset)$ is the unconditional prediction and $\epsilon_\theta(\cdot, C)$

is the conditional prediction. Many diffusion models, including text-to-image models [1, 8, 33, 34], employ this technique to achieve high-quality generation.

4. Method

4.1. Overview

Given a facial reference image $X_R \in \mathbb{R}^{H \times W \times 3}$ with height H and weight W , our objective is to generate a target facial image X_T that retains the identity and detailed attributes of X_R , while aligning with an image-aligned target control $D_T \in \mathbb{R}^{H \times W \times 9}$, which is defined by 3DMM renderings [9, 24] consisting of surface normal, albedo, and Lambertian renderings.

We employ dual branch U-Nets: the proposed FaceNet to capture the local and fine-grained appearance of X_R and a denoising U-Net for generation. FaceNet’s detailed features are injected into the denoising U-Net through the augmented self-attention layer. Additionally, we utilize CLIP [31] image encoder to provide coarse visual features to both U-Nets. To guide manipulation by control signal, we present a lightweight encoder called face controller, which embeds D_T directly into the denoising U-Net. However, as it only encodes D_T , the model cannot fully map the relation between X_T and X_R , limiting control adherence. Hence, we propose control mixer module (CMM) with multiple cross-attention layers to embed correlated features between D_T and D_R which is then fused into the augmented self-attention layer to guide the model’s attention effectively. D_R is another 3DMM renderings extracted from the reference image. Fig. 3 illustrates our overall architecture. For further control adherence, we employ reference control guidance (RCG) during inference. Each component is explained in details in the following sections.

4.2. Acquiring Paired Quadruplets for Face Rigging

As shown in Fig. 2, training our model with image datasets where $X_R = X_T$ like previous methods [7, 19] may lead the model to replicate X_R and ignore D_T . Hence, we leverage facial video dataset, CelebV-HQ [52], which contains 35,666 facial video clips involving 15,653 identities with various actions. From each video, we randomly select two frames, assigning one as X_R and the other as X_T . We then extract image-aligned surface normals, albedo, and Lambertian rendering from each frames. The extracted renderings from the reference image serve as the reference control D_R , while those from the target image serve as the target control D_T . Consequently, our dataset consists of a set of quadruplets consisting of $\{X_R, X_T, D_R, D_T\}$.

4.3. Injecting Appearance of Reference Image

To acquire strong image representations that preserve identity and fine details during generation, the reference image

50]。最近，ControlNet[50]通过将时间步特定控制特征整合到扩散U-Net中，实现了对深度、边缘和姿态等属性的精确控制。在面部编辑领域，方法[7、12、19、25]采用了类似的3DMM架构以实现显式面部控制。然而，这些方法往往难以从参考图像中编码精细细节。为克服这一局限，我们采用双分支结构：一个用于编码参考面部图像，另一个用于生成处理后的图像。

3. 前言

三维可变形面部模型（3DMMs）。3DMMs [2, 3, 24] 是一种面部参数模型，旨在通过紧凑的潜在火焰模型[24]精确表征面部姿态、形状和表情。作为广泛使用的3DMM之一，它以形状、姿态和表情作为输入，输出面部网格。本研究采用DECA [9]模型，该模型能够从面部图像中预测火焰[24]参数，同时获取正射相机参数、反照率和球谐照明参数。利用这些参数，我们可应用朗伯反射率获取表面法线、反照率和朗伯渲染。这些渲染结果提供像素级控制，包含适用于面部绑定的几何和纹理信息。因此，我们将这些渲染结果作为控制参数。**潜在扩散模型（LDM）。**LDM [34] 是一种生成模型，通过逐步学习去除VAE [21]潜在空间中的随机高斯噪声。在训练过程中，输入图像 X 通过VAE编码器被编码为潜在表示 z 。随后，该潜在表示 z 在随机选择的时间步 t 通过预定义的扩散前向过程受到高斯噪声 ϵ 的扰动，生成含噪潜在表示 z_t 。模型的核心目标是预测 ϵ ，即有效学习消除噪声。训练目标可表示为：

$$\mathcal{L}_{LDM} = \mathbb{E}_{z,t,\epsilon \sim \mathcal{N}(0,1)} [\|\epsilon_\theta(z_t; t) - \epsilon\|_2^2]. \quad (1)$$

无分类器引导（CFG）。为增强推理阶段扩散模型的条件生成能力，CFG [16]提出通过预测噪声引导生成过程，使其偏离无条件分布而趋近于条件分布。这使得我们能够从更精确的条件分布中采样，从而提升条件保真度。改进后的预测公式可表述如下：

$$\hat{\epsilon}_\theta(\cdot, C) = \epsilon_\theta(\cdot, \emptyset) + w(\epsilon_\theta(\cdot, C) - \epsilon_\theta(\cdot, \emptyset)), \quad (2)$$

其中 $\emptyset$ 表示无条件标签（如空文本）， C 表示条件输入， w 为引导尺度。因此 $\hat{h}_\theta(\cdot, \cdot)$ 是无条件预测， $\hat{h}_\theta(\cdot, C)$ 为

是条件预测。许多扩散模型，包括文本到图像模型[1, 8, 33, 34]，采用该技术以实现高质量生成。

4. 方法

4.1. 概述

给定一张面部参考图像 $X_R \in \mathbb{R}^{H \times W \times 3}$ ，其高度为 H 、权重为 W ，我们的目标是生成一张目标面部图像 X_T ，该图像需保留 X_R 的身份和细节特征，同时与图像对齐的目标控制图像 $D_T \in \mathbb{R}^{H \times W \times 9}$ 对齐，该控制图像由3DMM渲染[9, 24]定义，包含表面法线、反照率和朗伯反射渲染。

我们采用双分支U-Net架构：所提出的FaceNet用于捕捉 X_R 的局部与精细外观特征，而去噪U-Net则负责生成任务。FaceNet的详细特征通过增强自注意力层注入去噪U-Net。此外，我们利用CLIP[31]图像编码器为两个U-Net提供粗视觉特征。为引导控制信号的操作，我们提出一种轻量级编码器称为face controller，其将 D_T 直接嵌入去噪U-Net。但由于仅编码 D_T ，模型无法完全映射 X_T 与 X_R 之间的关系，导致控制依从性受限。因此，我们提出采用多交叉注意力层的控制混合模块（CMM），将 D_T 与 D_R 之间的相关特征进行嵌入，随后通过增强自注意力层融合以有效引导模型注意力。 D_R 是从参考图像中提取的另一组3DMM渲染结果。图3展示了我们的整体架构。为确保控制精度，我们在推理过程中采用参考控制引导（RCG）。各组件的详细说明见下文各章节。

4.2. 获取配对四元组进行面部绑定

如图2所示，使用 $X_R = X_T$ 的图像数据集训练模型（如先前方法[7, 19]）可能导致模型复制 X_R 而忽略 D_T 。因此，我们利用包含35,666段面部视频片段的CelebV-HQ[52]数据集，该数据集涉及15,653个身份的多种动作。从每个视频中随机选取两帧，分别指定一帧为 X_R ，另一帧为 X_T 。随后从每帧提取图像对齐的表面法线、反照率和朗伯渲染。参考图像的提取渲染作为参考控制 D_R ，目标图像的提取渲染作为目标控制 D_T 。因此，我们的数据集由一组四元组组成，包括 $\{X_R, X_T, D_R, D_T\}$ 。

4.3. 注入参考图像的外观

为在生成过程中获取既能保留图像特征又可保持细节的高质量图像表示，参考图像

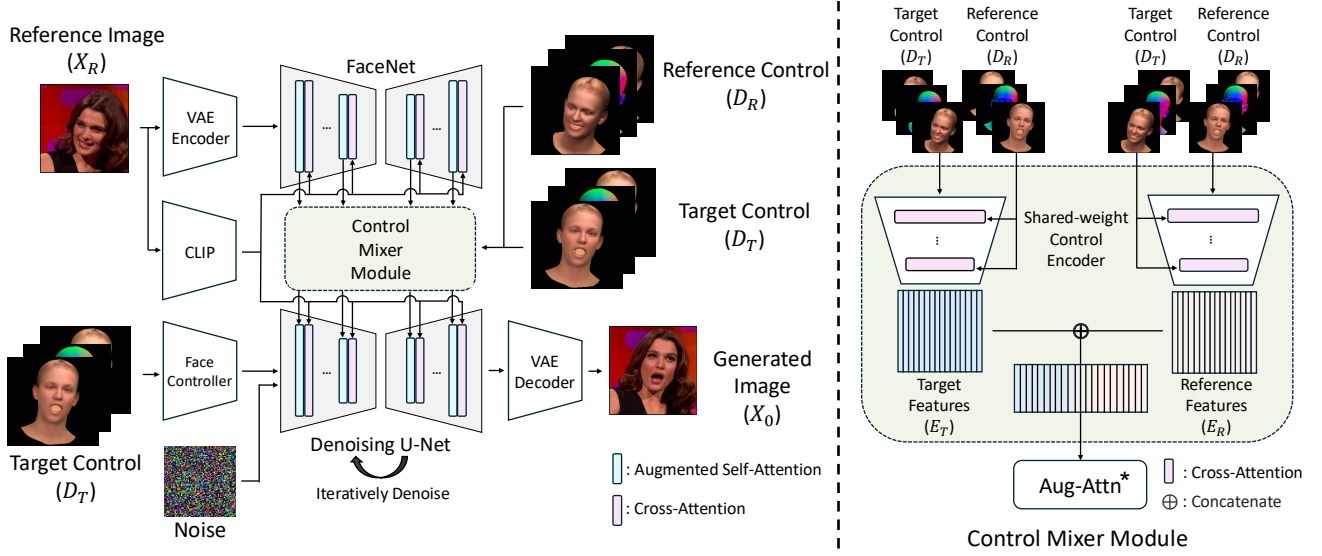


Figure 3. **Overall Architecture.** ControlFace encodes the reference image X_R into the FaceNet and CLIP image encoder for identity and semantic preservation. For face control, the target control D_T is incorporated into the denoising U-Net through face controller. To enhance the control adherence, the correlated feature between reference control D_R and target control D_T is acquired from the proposed control mixer module.

is encoded using both the CLIP [31] image encoder and the proposed FaceNet. The CLIP image embeddings, which capture high-level semantic information, are integrated into both denoising U-Net and FaceNet through cross-attention layers. Simultaneously, FaceNet encodes the reference facial image to extract detailed features which are then incorporated into the denoising U-Net through augmented self-attention mechanism [18, 37, 40, 47]. Since FaceNet and the denoising U-Net share the same architecture, they integrate seamlessly in self-attention by concatenating keys and values from both U-Nets and applying attention operation with queries from the denoising U-Net. Let $Q, K, V \in \mathbb{R}^{l \times d}$ denote query, key, value from the denoising U-Net where l and d denote length and dimensionality. Also, let $K^{\text{face}}, V^{\text{face}} \in \mathbb{R}^{l \times d}$ represent key, and value from the FaceNet, respectively. Then the augmented self-attention Aug-Attn is formulated as follows:

$$\text{Aug-Attn} = \text{Softmax}\left(\frac{Q[K, K^{\text{face}}]^T}{\sqrt{d}}\right)[V, V^{\text{face}}], \quad (3)$$

where $[\cdot, \cdot]$ denotes concatenation, and $\text{Softmax}(\cdot)$ means a softmax operation.

4.4. Incorporating Target Controls

Face Controller. To manipulate the reference image, we directly encode the target control D_T using a CNN-based shallow network which we denote face controller. The encoded features from the face controller are added to the first layer of the denoising U-Net. We have tested alternative diffusion conditioning methods, like ControlNet [50] and

ControlNeXt [29], but led to worse performance with increased computational complexity. The ablation study on the conditioning methods can be found in the Section 5.4.

Control Mixer Module. CMM outputs correlated embeddings for both controls D_R and D_T , denoted E_R , and E_T , respectively. CMM encodes these controls using two shared-weight control encoders. The control encoder consists of multiple convolution layers along with the cross-attention layers which is capable of modeling the interaction between the two control effectively. Specifically, when one control is input, the other serves as a conditioning input, passed through a convolution layer to match the resolution with the input.

The structure of CMM is illustrated in Fig. 3. The resulting embeddings E_R and E_T are in four different resolutions corresponding to the four different resolutions of U-Net activation maps. These embeddings are fused with features from the two U-Nets in the augmented self-attention layer. We modify Eq. 3 by adding E_R and E_T to the corresponding keys and values and denote it as Aug-Attn*. This can provide guidance on where the model should attend to. The modified augmented self-attention can be formulated as:

$$\begin{aligned} \text{Aug-Attn}^* = & \\ \text{Softmax}\left(\frac{(Q + E_T)[K + E_T, K^{\text{face}} + E_R]^T}{\sqrt{d}}\right)[V, V^{\text{face}}]. \end{aligned} \quad (4)$$

4.5. Training Objective

Since we perform the generation process in the latent space of a VAE [21], we encode X_R and X_T via the VAE encoder

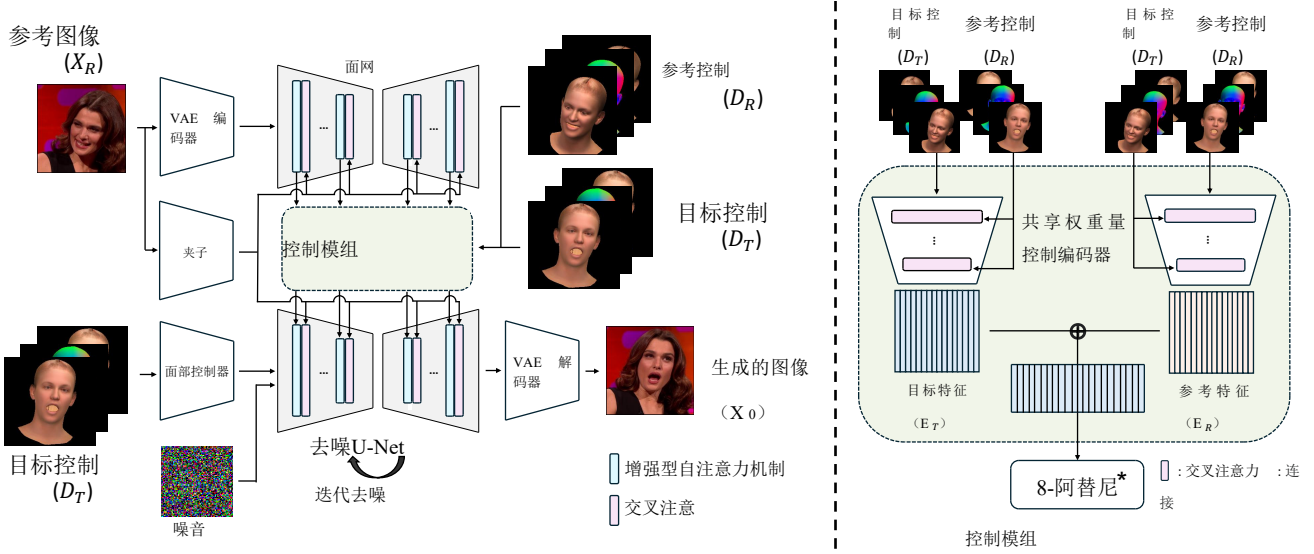


图3. **整体架构**。ControlFace将参考图像 X_R 编码至FaceNet和CLIP图像编码器，以保持身份和语义。对于人脸控制，目标控制 D_T 通过人脸控制器整合到去噪U-Net中。为增强控制一致性，参考控制 D_R 与目标控制 D_T 之间的相关特征通过所提出的控制混合模块获取。

采用CLIP[31]图像编码器和提出的FaceNet进行编码。捕捉高层次语义信息的CLIP图像嵌入通过交叉注意力层整合到去噪U-Net和FaceNet中。同时，FaceNet对参考人脸图像进行编码以提取详细特征，这些特征随后通过增强自注意力机制[18, 37, 40, 47]融入去噪U-Net。由于FaceNet与去噪U-Net共享相同架构，它们通过拼接两个U-Net的键值并应用来自去噪U-Net的查询进行注意力操作，从而在自注意力机制中无缝集成。设 $Q, K, V \in \mathbb{R}^{l \times d}$ 表示去噪U-Net的查询、键、值，其中 l 和 d 分别表示长度和维度。同时，设 $K^{\text{face}}, V^{\text{face}} \in \mathbb{R}^{l \times d}$ 分别表示FaceNet的键和值。随后，增强型自注意力机制Aug-Attn的数学表达式如下：

$$\text{Aug-Attn} = \text{Softmax}\left(\frac{Q[K, K^{\text{face}}]^T}{\sqrt{d}}\right)[V, V^{\text{face}}], \quad (3)$$

其中 $[\cdot, \cdot]$ 表示拼接， $\text{Softmax}(\cdot)$ 表示softmax运算。

4.4. 纳入目标控制

面部控制器。为操纵参考图像，我们直接使用基于CNN的浅层网络（记为面部控制器）对目标控制 D_T 进行编码。来自面部控制器的编码特征被添加到去噪U-Net的第一层。我们测试了其他扩散条件方法，例如ControlNet[50]和

ControlNeXt [29]，但随着计算复杂度的增加导致性能下降。关于预处理方法的消融研究可参见第5.4节。

控制混合器模块。CMM为两个控制变量 D_R 和 D_T 分别输出相关嵌入向量，记为 E_R 和 E_T 。CMM使用两个共享权重的控制编码器对这些控制变量进行编码。控制编码器由多个卷积层和交叉注意力层组成，能够有效建模两个控制变量之间的交互关系。具体而言，当输入一个控制变量时，另一个作为条件输入，通过卷积层进行匹配以适应输入分辨率。

CMM的结构如图3所示。生成的嵌入 E_R 和 E_T 对应于U-Net激活图的四种不同分辨率。这些嵌入在增强的自注意力层中与两个U-Net的特征融合。我们通过在相应的键和值中添加 E_R 和 E_T 来修改等式3，并将其表示为Aug-Attn*。这可以为模型应关注的位置提供指导。修改后的增强自注意力可以表述为：

$$\begin{aligned} \text{Aug-Attn}^* = & \text{Softmax}\left(\frac{(Q + E_T)[K + E_T, K^{\text{face}} + E_R]^T}{\sqrt{d}}\right)[V, V^{\text{face}}]. \end{aligned} \quad (4)$$

4.5. 培训目标

由于我们在VAE的潜在空间中执行生成过程[21]，我们通过VAE编码器对 X_R 和 X_T 进行编码

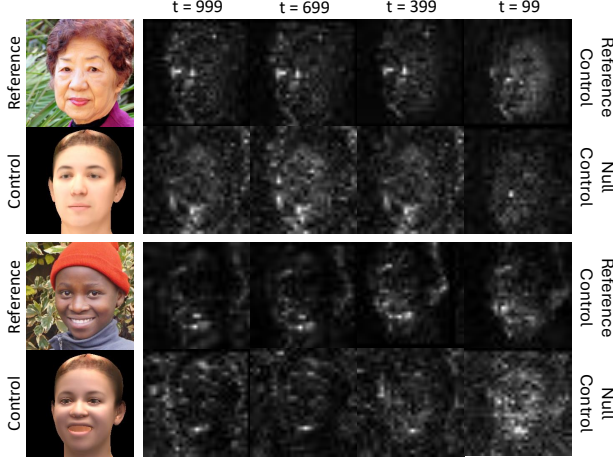


Figure 4. **Visualization of Reference Control Guidance.** We visualize the deltas, $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, \emptyset)$ and $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, D_R)$, which corresponds to CFG [16] applied to face controller input and RCG, respectively, across different timesteps t . The first and third row display RCG deltas whereas second and fourth row show the CFG deltas. The former shows noisy deltas over all the timesteps.

and acquire latents z_R and z_T , respectively. Our model $\epsilon_\theta(\cdot)$ takes z_T, z_R, D_T , and D_R as input and learns to predict random Gaussian Noise ϵ which is used to perturb z_R . Similar to Eq. 1, our training objective is as follows:

$$\mathcal{L} = \mathbb{E}_{z, t, \epsilon \sim \mathcal{N}(0, 1)} [\|\epsilon_\theta(z_T, t; z_R, D_T, D_R) - \epsilon\|_2^2], \quad (5)$$

where $z_{R, t}$ is the perturbed z_R with predefined diffusion forward schedule with randomly selected timestep t .

4.6. Reference Control Guidance

At inference, we further enhance the control adherence with a novel guidance technique. To achieve this, we first naively apply CFG to face controller input. This can be formulated as follows:

$$\hat{\epsilon}_\theta(\cdot, D_T) = \epsilon_\theta(\cdot, \emptyset) + w(\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, \emptyset)), \quad (6)$$

where $\emptyset$ is a null input for the face controller, similar to Eq. 2. However, Tab. 5 shows that the improvement is negligible compared to the result without any guidance. We suspect that this is because the null condition cannot provide a good grounding for where to deviate away from. Therefore, our proposed guidance method, RCG, utilizes D_R instead of null conditioning for better grounding. We define it as follows:

$$\hat{\epsilon}_\theta(\cdot, D_T) = \epsilon_\theta(\cdot, D_R) + w(\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, D_R)). \quad (7)$$

In Fig. 4, we visualize each difference, $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, \emptyset)$ and $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, D_R)$, on various timesteps. Although the difference of both guidances tends to focus on facial area on lower timesteps, the former is noisy over

all timesteps which can compromise the generation process. On the other hand, the difference of RCG shows relatively clean face-aligned estimates over all timesteps. Hence, we leverage Eq. 7 for generating our output X_0 .

5. Experiments

5.1. Implementation Details

We train our model on CelebV-HQ [52] dataset, resizing all frames to 256×256 pixels. We extract 3DMM renderings also of shape 256×256 from each frame leveraging DECA [9]. The weights of the FaceNet and the denoising U-Net are first initialized from a variant of pretrained Stable Diffusion v1.5 [34], which is a large text-to-image model fine-tuned to accept CLIP [31] image embeddings as input. We present more implementation details in the Appendix ??.

5.2. Comparison

Baselines. We compare our method with four publicly available 3DMM-controllable face generation methods: GIF [10], CapHuman [25], Arc2Face [28], and DiffusionRig [7]. GIF is a 3DMM-conditional GAN-based approach utilizing the StyleGAN [20] generator, but it does not take a reference image as input. Other methods are diffusion-based. CapHuman and Arc2Face utilize a face recognition model [5] feature for identity preservation, whereas DiffusionRig employs ResNet [14] to encode the reference image features. CapHuman also accepts text input for additional control; therefore, we fix the input text to “a photo of a person” in our experiments to ensure a fair comparison.

We evaluate ControlFace and the baselines on the FFHQ [20] dataset, which consists of over 70,000 high-quality facial images. Notably, all baselines except CapHuman were trained on FFHQ, whereas our method was not. Additionally, ControlFace is trained on CelebV-HQ [52], which contains approximately 15,000 unique identities—fewer than FFHQ. This demonstrates our method’s robustness and generalizable capability.

Qualitative Comparison. We compare our generated results with the baselines on three identities from the FFHQ dataset. Fig. 5 shows the results. We present rigged outputs with variations in three attributes—pose, expression, and lighting—for all methods except Arc2Face, as it does not support lighting control. For visualization, we display only the Lambertian rendering of the control.

While all methods produce results aligned with the control parameters, GIF fails to preserve identity as it does not incorporate any image-related features. CapHuman and Arc2Face maintain the identity but lose semantic details like hairstyle and background since they utilize only facial features. DiffusionRig shows good performance in identity and detail preservation; however, the identity tends to change

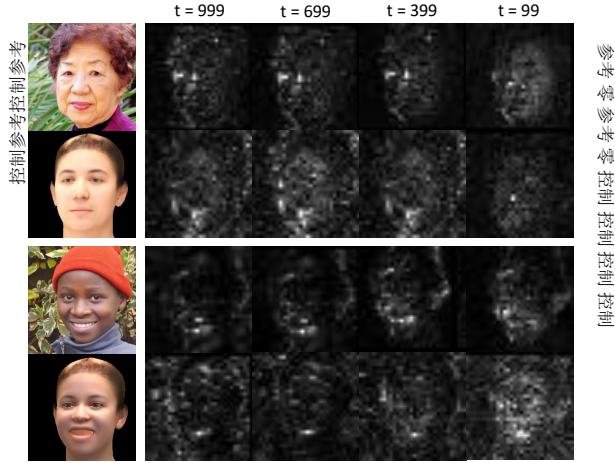


图4. 参考控制引导的可视化。我们可视化了增量， $h_{\theta}(\cdot, D_T) - h_{\theta}(\cdot, \cdot)$ 和 $h_{\theta}(\cdot, D_T) - h_{\theta}(\cdot, D_R)$ ，这分别对应于应用于面部控制器输入和 RCG 的 CFG [16] 在不同时间步 t 上的表现。第一行和第三行显示 RCG 增量，而第二行和第四行显示 CFG 增量。前者展示了所有时间步上的噪声增量。

并分别获取潜在变量 z_R 和 z_T 。我们的模型 $h_{\theta}(\cdot)$ 以 z_T, z_R, D_T 和 D_R 作为输入，并学习预测用于扰动 z_R 的随机高斯噪声 h 。类似于等式1，我们的训练目标如下：

$$\mathcal{L} = \mathbb{E}_{z, t, \epsilon \sim \mathcal{N}(0, 1)} [\|\epsilon_{\theta}(z_T, t; z_R, D_T, D_R) - \epsilon\|_2^2], \quad (5)$$

其中 $z_{R, t}$ 是具有预定义扩散前向调度且随机选择时间步 t 的扰动 z_R 。

4.6. 参考控制指南

在推理阶段，我们通过一种创新的引导技术进一步提升控制依从性。为此，我们首先将 CFG 直接应用于面部控制器输入。具体可表述为：

$$\hat{\epsilon}_{\theta}(\cdot, D_T) = \epsilon_{\theta}(\cdot, \emptyset) + w(\epsilon_{\theta}(\cdot, D_T) - \epsilon_{\theta}(\cdot, \emptyset)), \quad (6)$$

其中 $\emptyset$ 是面部控制器的无效输入，类似于等式2。然而，表5显示，与没有任何引导的结果相比，改进效果微乎其微。我们推测这是因为无效条件无法为偏离提供良好的基准。因此，我们提出的引导方法 RCG 利用 D_R 而非无效条件来获得更好的基准。我们将其定义如下：

$$\hat{\epsilon}_{\theta}(\cdot, D_T) = \epsilon_{\theta}(\cdot, D_R) + w(\epsilon_{\theta}(\cdot, D_T) - \epsilon_{\theta}(\cdot, D_R)). \quad (7)$$

在图4中，我们可视化了各时间步长上的差异： $h_{\theta}(\cdot, D_T) - h_{\theta}(\cdot, \cdot)$ 与 $h_{\theta}(\cdot, D_T) - h_{\theta}(\cdot, D_R)$ 。尽管两种引导方式的差异在较低时间步长上倾向于聚焦面部区域，但前者在

所有可能影响生成过程的时间步。另一方面，RCG 的差异在所有时间步上显示出相对干净的面部对齐估计。因此，我们利用等式7来生成我们的输出 X_0 。

5. 实验

5.1. 实施细节

我们在 CelebV-HQ[52] 数据集上训练模型，将所有帧调整为 256×256 像素。利用 DECA[9] 从每帧提取形状为 256×256 的 3DMM 渲染图。FaceNet 和去噪 U-Net 的权重首先从预训练的 Stable Diffusion v1.5[34] 的变体初始化，该模型是经过微调以接受 CLIP[31] 图像嵌入作为输入的大型文本到图像模型。更多实现细节详见附录??。

5.2. 比较

基线方法。我们将本方法与四种公开可用的 3DMM 可控人脸生成方法进行比较：GIF[10]、CapHuman[25]、Arc2Face[28] 和 DiffusionRig[7]。GIF 是一种基于 StyleGAN[20] 生成器的 3DMM 条件 GAN 方法，但不接受参考图像作为输入。其他方法基于扩散机制。CapHuman 和 Arc2Face 利用人脸识别模型[5]特征进行身份保留，而 DiffusionRig 采用 ResNet[14] 对参考图像特征进行编码。CapHuman 还接受文本输入以实现额外控制；因此我们在实验中将输入文本固定为“一张人物照片”以确保公平比较。

我们在 FFHQ[20] 数据集上评估了 ControlFace 和基线方法，该数据集包含超过 70,000 张高质量面部图像。值得注意的是，除 CapHuman 外，所有基线方法均基于 FFHQ 训练，而我们的方法则不然。此外，ControlFace 是在 CelebV-HQ[52] 数据集上训练的，该数据集包含约 15,000 个独特身份——少于 FFHQ。这证明了我们方法的鲁棒性和泛化能力。

定性对比。我们通过 FFHQ 数据集集中的三个身份特征，将生成结果与基准模型进行对比。图5展示了对比结果。除 Arc2Face（该方法不支持光照控制）外，我们展示了所有方法在姿态、表情和光照三个属性上进行调整的伪造输出。为便于可视化，仅呈现对照组的朗伯光照渲染效果。

尽管所有方法生成的结果均符合控制参数，但 GIF 因未整合图像特征而无法保持身份一致性。CapHuman 和 Arc2Face 虽能保留身份特征，却因仅使用面部特征而丢失发型、背景等语义细节。DiffusionRig 在身份与细节保留方面表现优异，但其身份特征易发生改变。



Figure 5. **Qualitative Results.** We compare the results of rigging pose, expression, and light with four different baselines [7, 10, 25, 28]. The reference images are from the FFHQ [20] dataset. Compared to the baselines ControlFace aligns with the target control, maintaining the identity and details in the reference image.

when the control differ significantly from those of the reference image. Compared to all baselines, our method exhibits superior performance in both identity and detail preservation while adhering to the target controls. We present more generated results on FFHQ dataset in the Appendix D.1.

Moreover, since DiffusionRig originally requires additional training on a personal album for optimal performance, we fine-tune it on the reference image. As shown in Fig. 6, while fine-tuning improves DiffusionRig’s identity and detail preservation, it still struggles with identity shifts under large pose changes. In contrast, our method maintains consistent identity even under large pose variations, demonstrating robustness and fidelity in generated results.

Quantitative Comparison. We measure the DECA re-inference error using the same settings as previous works [7, 10, 19] to evaluate fidelity to the target controls. Specifi-

cally, we generate 1,024 images by varying shape, lighting, expression, and pose parameters. For each generated image, we use DECA to re-inference the 3DMM parameters. We then calculate the Root Mean Square Error (RMSE) between the FLAME mesh rendered with original control parameters and the newly inferred ones. For lighting, we compute the RMSE directly on the spherical harmonics coefficients. Tab. 1 shows that our method achieves the best control adherence on average.

We evaluate image quality using three metrics: identity preservation (ID) score via cosine similarity of features from a face recognition model [5], Fréchet Inception Distance (FID) [15] for image quality, and Learned Perceptual Image Patch Similarity (LPIPS) [51] for semantic consistency. These metrics are computed on the same set of 4096 images used for DECA re-inference error evaluation. ID and LPIPS are not calculated for GIF, as it does not use



图5. 定性结果。我们将骨骼姿态、表情和光照的结果与四种不同的基线方法[7、10、25、28]进行比较。参考图像来自 FFHQ [20]数据集。与基线方法相比，ControlFace与目标对照保持对齐，保留了参考图像中的身份和细节。

当控制图像与参考图像存在显著差异时。相较于所有基准方法，我们的方法在保持目标控制的同时，展现出在图像身份识别和细节保留方面的卓越性能。我们将在附录D.1中展示更多 FFHQ 数据集上的生成结果。

此外，由于DiffusionRig最初需要在个人相册上进行额外训练才能达到最佳效果，我们选择在参考图像上进行微调。如图6所示，虽然微调提升了DiffusionRig的身份识别和细节保留能力，但在大角度姿态变化时仍存在身份偏移问题。相比之下，我们的方法即使在大角度姿态变化下也能保持身份一致性，展现出生成结果的稳健性和保真度。

定量比较。 我们采用与先前研究[7、10、19]相同的参数设置来测量DECA再推理误差，以此评估其对目标对照的保真度。具体-

具体来说，我们通过调整形状、光照、表情和姿态参数生成了1,024张图像。针对每张生成图像，我们使用DECA算法重新推断3DMM参数。随后计算原始控制参数渲染的火焰网格与新推断参数之间的均方根误差（RMSE）。在光照参数方面，我们直接对球谐函数系数进行 RMSE 计算。表1显示，我们的方法平均实现了最佳的控制一致性。

我们采用三项指标评估图像质量：通过人脸识别模型特征的余弦相似度计算的身份保留度（ID）评分[5]、Fréchet-伊辛普森距离（FID）[15]用于评估图像质量，学习感知图像块相似度（LPIPS）[51]用于评估语义一致性。这些指标均基于用于DECA再推理误差评估的4096张图像集计算得出。由于GIF格式不使用该指标，因此未计算其ID和LPIPS。

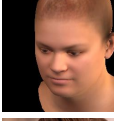
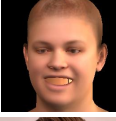
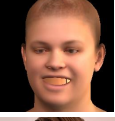
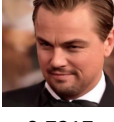
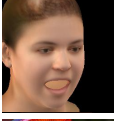
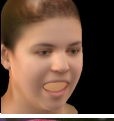
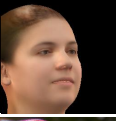
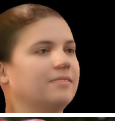
	Ours (Zero-shot)	DiffusionRig (Finetuned)	Ours (Zero-shot)	DiffusionRig (Finetuned)
Reference				
Control				
Results				
ID ↑	0.7217	0.7085	0.8252	0.7822
Re-Infer. ↓	10.31	16.36	2.83	2.94
Reference				
Control				
Results				
ID ↑	0.5850	0.5161	0.7852	0.7515
Re-Infer. ↓	13.42	22.59	2.58	2.27

Figure 6. **Comparison with Fine-tuning Method.** We compare the results with the fine-tuned DiffusionRig [7] on two reference images. We show the generated results along with the identity similarity (ID) [5] and DECA [9] re-inference error.

a reference image. Results in Tab. 2 show our method achieves the lowest FID and LPIPS, indicating high image quality and semantic consistency. While Arc2Face scores highest in identity preservation due to explicit facial feature conditioning, its high LPIPS score indicates a lack of background preservation. In contrast, our method not only preserves identity but also maintains semantic consistency with the reference image, offering a balanced performance across all evaluated metrics. Also, for a fair comparison, we train DiffusionRig on CelebV-HQ and apply the same evaluation metrics which is illustrated in the Appendix C.1.

We conduct a user study for additional evaluation, following ImagenHub standards [22]. Participants rate each model on semantic consistency (SC), measuring alignment between the condition and the generated image, and perceptual quality (PQ), assessing image quality. We compare our model with CapHuman and DiffusionRig, both of which use reference images and lighting as conditions. Tab.3 presents

Table 1. **Quantitative Results on Control Adherence.** We compare the DECA [9] re-inference errors following previous work [10] with four different baselines [7, 10, 25, 28]. Our method, ControlFace, shows the best control adherence on average.

	Light ↓	Shape ↓	Exp. ↓	Pose ↓	Avg. ↓
GIF [10]	17.04	2.29	8.16	8.17	8.91
CapHuman [25]	15.16	2.65	6.68	19.03	10.40
Arc2Face [5]	-	3.13	13.83	15.37	10.77
DiffusionRig [7]	6.31	2.11	5.58	6.26	5.06
ControlFace (Ours)	3.75	2.56	5.43	7.67	4.85

Table 2. **Quantitative Results on Identity Preservation and Image Quality.** We evaluate identity similarity (ID) [5] and LPIPS [51] to assess appearance preservation of the reference image, and FID [15] to measure overall image quality.

	ID ↑	FID ↓	LPIPS ↓
GIF [10]	-	24.48	-
CapHuman [25]	0.4356	59.78	0.4556
Arc2Face [5]	0.7825	17.82	0.5253
DiffusionRig [7]	0.2042	23.05	0.3758
ControlFace (Ours)	0.7586	15.50	0.1429

Table 3. **User Study.** We ask the participants to evaluate the model on condition alignment (SC) and perceptual quality (PQ). Our model achieves the highest score on all metrics.

	SC ↑	PQ ↑	OV ↑
CapHuman [25]	0.4086	0.7019	0.4545
DiffusionRig [7]	0.3533	0.5072	0.3473
ControlFace (Ours)	0.8750	0.8605	0.8486

SC, PQ, and the overall score (OV), which is the geometric mean of SC and PQ demonstrating the final score of the model. Our model achieves the highest scores across all metrics. Further details of the user study is available in the Appendix B.

5.3. Result on Out-of-Domain images

To further evaluate the zero-shot generalization capability of our model, we test ControlFace on out-of-domain reference images that significantly differ from the real face data leveraged during training. Specifically, we employ animation-like facial images as reference images. As shown in Fig. 7, our model successfully generates rigged results for these animation faces, demonstrating that it can effectively handle outside the scope of its training data. We present more examples in the Appendix D.2.

5.4. Ablation Studies

Scaling Reference Control Guidance. We examine the effect of guidance scales w on control adherence. From here, CFG* and CFG[†] refer to CFG applied to the face controller and CMM, respectively, both incorporating null inputs. On the other hand, CFG represents applying a null

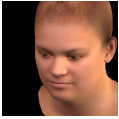
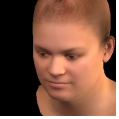
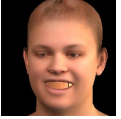
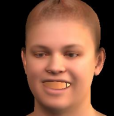
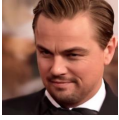
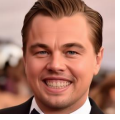
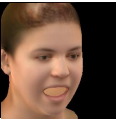
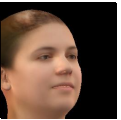
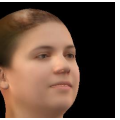
	我们的 (零示例)	DiffusionRig (Finetuned)	我们的 (零示例)	DiffusionRig (Finetuned)
参考				
统治				
结果				
ID ↑ 重新 推断。 ↓	0.7217 10.31	0.7085 16.36	0.8252 2.83	0.7822 2.94
参考				
统治				
结果				
ID ↑ 重新 推断。 ↓	0.5850 13.42	0.5161 22.59	0.7852 2.58	0.7515 2.27

图6.与微调方法的比较。我们在两张参考图像上将结果与微调后的DiffusionRig[7]进行比较。我们展示了生成结果以及身份相似度 (ID) [5]和DECA[9]的再推理误差。

参考图像。表2的实验结果表明，我们的方法在FID和 LPIPS 指标上表现最优，这充分体现了图像质量与语义一致性的双重优势。虽然Arc2Face凭借显式面部特征条件化在身份保留方面表现突出，但其 LPIPS 评分偏高，反映出背景信息的保留不足。相比之下，我们的方法不仅精准还原了人物身份特征，还与参考图像保持了语义一致性，从而在各项评估指标上实现了性能平衡。为确保公平性，我们在 CelebV-HQ数据集上训练了DiffusionRig模型，并采用附录C.1所示的相同评估标准进行验证。

我们按照ImagenHub标准[22]开展用户研究以进行补充评估。参与者需对每个模型进行语义一致性 (SC) 评分 (衡量条件与生成图像的匹配程度) 和感知质量 (PQ) 评分 (评估图像质量)。我们将模型与CapHuman和DiffusionRig进行对比，这两者均采用参考图像和光照作为条件。表3展示了

表1.控制依从性定量结果。我们比较了DECA[9]在先前情况下的再推理错误 [10] 与四种不同基线[7, 10, 25, 28]进行对比。我们的方法ControlFace在平均控制依从性方面表现最佳。

	光 ↓	形状 ↓	体验 ↓	姿态 ↓	Avg. ↓
GIF [10]	17.04	2.29	8.16	8.17	8.91
CapHuman [25]	15.16	2.65	6.68	19.03	10.40
Arc2Face [5]	-	3.13	13.83	15.37	10.77
DiffusionRig [7]	6.31	2.11	5.58	6.26	5.06
控制面	3.75	2.56	5.43	7.67	4.85

表2.身份保持与图像质量的定量结果。我们评估了身份相似度 (ID) [5]和 LPIPS [51]以评估参考图像的外观保持情况，并评估了FID[15]以衡量整体图像质量。

	ID ↑	FID ↓	LPIPS ↓
GIF [10] CapHu-	-	24.48	-
man [25] Arc2Face	0.4356	59.78	0.4556
[5] DiffusionRig	0.7825	17.82	0.5253
[7]	0.2042	23.05	0.3758
ControlFace (我们的)	0.7586	15.50	0.1429

表3.用户研究。我们要求参与者对模型在条件一致性 (SC) 和感知质量 (PQ) 方面的表现进行评估。我们的模型在所有指标上均获得最高分。

	SC ↑	PQ ↑	OV ↑
CapHuman [25]	0.4086	0.7019	0.4545
DiffusionRig [7]	0.3533	0.5072	0.3473
控制面	0.8750	0.8605	0.8486

SC、PQ及总分 (OV) ——后者为SC与PQ的几何平均数，用以体现模型的最终得分。本模型在所有指标上均取得最高分。用户研究的更多细节详见附录B。

5.3. 域外图像处理结果

为深入评估模型的零样本泛化能力，我们在训练时使用的真实人脸数据之外，对ControlFace模型进行了跨域测试。具体采用动画风格的人脸图像作为参考样本。如图7所示，模型成功生成了这些动画人脸的拟人化结果，证明其能有效处理超出训练数据范围的场景。更多案例详见附录D.2。

5.4. 消融研究

规模参考控制指导。我们研究了指导尺度 w 对控制依从性的影响。在此基础上， CFG^* 和 $CFG^\dagger$ 分别指应用于面部控制器和CMM的CFG，两者均包含零输入。另一方面，CFG表示应用一个零

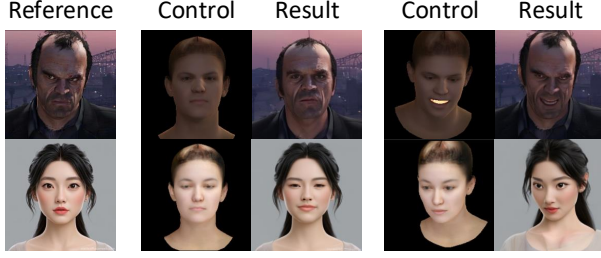


Figure 7. **Results on Out-of-domain Data.** We test ControlFace on animation-like faces to illustrate the robustness and generalizability.

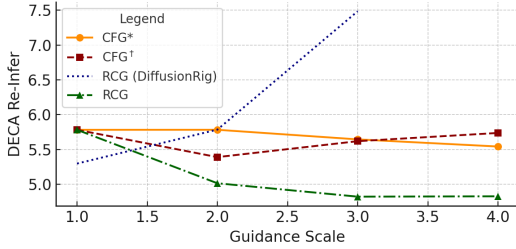


Figure 8. **Ablation on Guidance Scale.** We plot the DECA [9] re-inference error for guidance scales w from 1.0 to 4.0. Increasing w improves errors for RCG, CFG^* , and $\text{CFG}^\dagger$ to some extents, but greatly worsens for DiffusionRig [7].

input to the CLIP encoder, following the approach used in most text-to-image models [8, 33, 34]. Since RCG requires no null condition training, we also apply it to DiffusionRig, evaluating on 128 randomly selected FFHQ reference images. Fig. 8 shows that increasing the scale improves DECA re-inference error for RCG, CFG^* , and $\text{CFG}^\dagger$ to varying extents, but significantly reduces performance for DiffusionRig. We suspect this is due to DiffusionRig’s struggle to maintain identity consistency under different controls, causing conflicting identities and poor alignment when using both target and reference controls.

Ablation on Different Conditioning Modules. We attach different kinds of widely used conditioning modules [29, 50] to see their effects on control adherence. On top of FaceNet architecture, without CMM, we attach various conditioning modules, applying CFG during inference. Tab. 4 compare face controller with ControlNet [50] and ControlNeXt [29] showing the number of parameters and DECA re-inference error. Face controller performs the best while having the least number of trainable parameters.

Ablation on Architecture. Starting with FaceNet and Face Controller, we sequentially add CMM, CFG [16], CFG^* , and our proposed RCG, with results in Tab. 5. Incorporating CMM improves alignment and image quality but slightly reduces identity similarity compared to FaceNet. This occurs because enhanced control adherence requires adjustments to attributes like expression, pose, or lighting, which may inadvertently alter identity-related features.

Table 4. **Conditioning Module Ablation.** We plot parameter counts and DECA [9] re-inference errors for various conditioning modules.

	# of params.	Re-Infer.
ControlNet [50]	$\sim 360\text{M}$	8.37
ControlNeXt [29]	$\sim 3\text{M}$	8.06
Face controller (Ours)	$\sim 1\text{M}$	7.87

Table 5. **Architecture Ablation.** We plot DECA [9] re-inference error, identity similarity score [5], FID [15], and LPIPS [51] for each architecture.

	Component	Re-Infer.↓	ID↑	FID↓	LPIPS↓
(I)	FaceNet	7.13	0.8234	32.45	0.1321
(II)	(I) + CMM	5.78	0.7520	15.35	0.1384
(III)	(II) + CFG	5.77	0.7586	15.50	0.1429
(IV)	(II) + CFG^*	5.75	0.7308	15.43	0.1400
(V)	(II) + RCG (Ours)	4.85	0.7586	15.50	0.1429

Consequently, FaceNet, with poorer alignment, achieves the highest identity preservation. Our proposed RCG method compensates for the marginal decrease in ID score by achieving better control adherence and image quality than FaceNet. Also, RCG exhibits better control alignment among other guidance methods without sacrificing the identity. We additionally provide ablation study on different inputs and architecture for CCM in the Appendix C.3.

5.5. Limitation

ControlFace relies on DECA [9] for acquiring 3DMM renderings which may not capture the finer facial details needed for optimal performance. Adopting a more advanced 3D face estimation model [4, 54] could enhance the model’s overall performance. Additionally, ControlFace is currently trained on a single facial video dataset [52], limiting its exposure to a variety conditions and identities. Incorporating additional video datasets, as well as multiview facial datasets, could further improve the model’s generalizability.

6. Conclusion

In this paper, we introduce ControlFace, a face rigging method that addresses limitations arising from reliance on image datasets, such as loss of fine details and need for fine-tuning. Our dual-branch U-Net is capable of capturing the identity and intricate details of the reference image. To improve control adherence, we leverage CMM which learns embedding the relationship between the target and reference controls, and RCG which utilizes reference control for better grounding. Training on a facial video dataset improved identity and detail preservation. Experiments and ablations confirm ControlFace’s superior performance in identity consistency, control adherence, and image quality, showing its practicality for real-world applications.

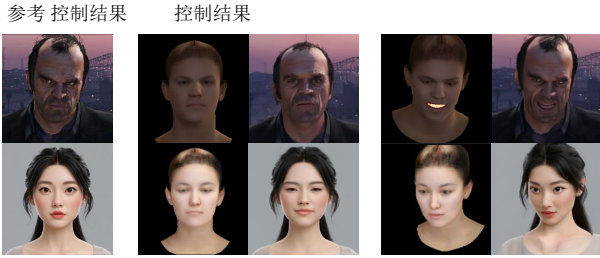


图7. 域外数据测试结果。我们在类动画人脸数据上测试ControlFace，以展示其鲁棒性和泛化能力。

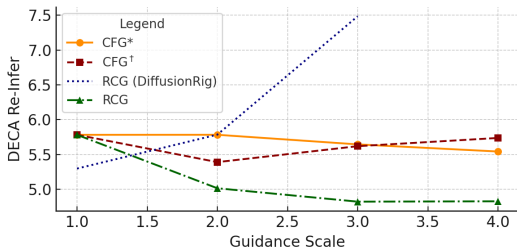


图8. 指导尺度上的消融实验。我们绘制了DECA[9]在指导尺度 w 从1.0到4.0时的再推理误差。增加 w 在一定程度上改善了RCG、CFG*和CFG†的误差，但对DiffusionRig[7]的误差却大幅恶化。

输入到CLIP编码器，采用大多数文本到图像模型[8, 33, 34]中使用的方法。由于RCG不需要零条件训练，我们也将其应用于DiffusionRig，并在128张随机选择的FFHQ参考图像上进行评估。图8显示，增加尺度会不同程度地提高RCG、CFG*和CFG†的DECA再推理误差，但显著降低DiffusionRig的性能。我们推测这是由于DiffusionRig在不同控制下难以保持身份一致性，导致在使用目标和参考控制时出现身份冲突和对齐不良。

不同条件模块的消融实验。我们将多种常用条件模块[29, 50]附加到FaceNet架构上，观察其对控制依从性的影响。在不使用CMM的情况下，我们在推理阶段应用CFG附加各种条件模块。表4对比了Face Controller与ControlNet[50]、ControlNeXt[29]，展示了参数数量和DECA再推理误差。Face Controller在可训练参数数量最少的情况下表现最佳。

架构消融实验。从FaceNet和Face Controller开始，我们依次添加CMM、CFG[16]、CFG*以及我们提出的RCG，结果如表5所示。与FaceNet相比，加入CMM虽能提升对齐效果和图像质量，但会略微降低身份相似度。这是因为增强的控制依从性需要调整表情、姿态或光照等属性，这些调整可能无意中改变身份相关特征。

表4. 条件模块消融。我们绘制了不同条件模块的参数计数和DECA[9]再推理错误。

	参数数量。重新推断。	
ControlNet [50]	~360M	8.37
ControlNeXt [29]	~3M	8.06
面控器 (Ours)	~1M	7.87

表5. 架构消融。我们绘制了DECA[9]再推理误差、身份相似度分数[5]、FID[15]以及LPIPS[51]在各架构中的表现。

	组件	重新推断。↓	ID↑	FID↓	LPIPS
(I)	面网	7.13	0.8234	32.45	0.1321
微光	(I) + CMM	5.78	0.7520	15.35	0.1384
罗马数字3	(II) + CFG	5.77	0.7586	15.50	0.1429
增值	(II) + CFG*	5.75	0.7308	15.43	0.1400
(V)	(II) + RCG (我们的)	4.85	0.7586	15.50	0.1429

因此，虽然FaceNet的对齐效果较差，但其身份保留率最高。我们提出的RCG方法通过比FaceNet更好的控制依从性和图像质量，弥补了身份识别分数的微小下降。此外，RCG在不牺牲身份识别的前提下，展现出优于其他引导方法的控制对齐效果。我们还在附录C.3中提供了CCM在不同输入和架构下的消融研究。

5.5. 限制

ControlFace依赖DECA[9]获取3DMM渲染图像，但该方法可能无法捕捉到实现最佳性能所需的精细面部细节。采用更先进的3D面部估计模型[4, 54]可提升模型整体性能。此外，ControlFace当前仅基于单一面部视频数据集[52]进行训练，导致其暴露于多样化场景和身份特征的范围有限。通过整合更多视频数据集及多视角面部数据集，可进一步增强模型的泛化能力。

6. 结论

本文提出ControlFace面部绑定方法，该方法突破了传统图像数据集的局限性，有效解决了细节丢失和微调需求等问题。我们设计的双分支U-Net网络能够精准捕捉参考图像的身份特征与精细细节。为提升控制贴合度，系统采用CMM学习目标控制与参考控制的嵌入关系，同时RCG利用参考控制进行更精准的定位。通过面部视频数据集训练，显著提升了身份识别与细节保留效果。实验与消融分析表明，ControlFace在身份一致性、控制贴合度及图像质量方面均表现优异，充分验证了其在实际应用中的可行性。